

Spatial Patterns in Educational Attainment Across Kentucky

Research Progress Report | Week 2

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1 Introduction

Kentucky has grown more educated over the past three decades, but we do not really yet know whether every part of the state share in it, or did the gains pile up in a few places while others stayed behind? This project answers that question by tracking educational attainment at the neighborhood level across Kentucky from 1990 to 2024, and asking whether the gap between high- and low-attainment places has widened or narrowed over time.

The unit of analysis is the census tract, a small area roughly the size of a neighborhood, harmonized to 2020 boundaries so that the same patch of ground means the same thing in every year (1,306 tracts). The attainment measure is the share of adults 25 and older who hold a bachelor's degree or higher, and it is built from the decennial census long form for 1990 and 2000 and from the American Community Survey for 2012 through 2024.

The central finding, presented across the sections that follow, is a tension between two things that are both true. Average attainment rose across the state, and the lowest-attainment neighborhoods improved the most in proportional terms. Yet the raw gap between the most- and least-educated places widened the whole way, and that widening is increasingly a story of whole counties pulling apart from one another. This report walks through how the dataset was rebuilt, what the maps and dispersion measures show, and where the divergence are.

2 Data

The question we ask in this project is about places, so the data has to describe places. The unit is the census tract, a small area about the size of a neighborhood, and the measure is the share of adults 25 and older who hold a bachelor's degree or higher. I track that share for every Kentucky

tract from 1990 to 2024, with all years put on 2020 tract boundaries so that each tract points to the same piece of ground across time (1,306 tracts in total).

The data come from two sources. For 2012 through 2024 I use the American Community Survey, table B15003, pulled through the `tidycensus` package in R. For 1990 and 2000 I use the decennial census long form from NHGIS. I pull and clean everything in R because `tidycensus` connects directly to the Census API and labels the variables on the way in. The pipeline I inherited had been built on hand-coded API calls that were brittle and hard to follow, which is part of why I rebuilt it from scratch rather than patching it.

BA+ is the sum of four lines in table B15003, bachelor's, master's, professional, and doctorate degrees, divided by the population 25 and over. This is where my numbers part ways with the inherited dataset. Natalie's BA+ ran about 11 to 14 percent; mine runs about 20 to 26. The likely reason is that hers counted only the bachelor's line and left out the three graduate degrees. Kentucky's published statewide share sits around 22 to 27 percent, which lines up with mine, so the definition the project uses is the one that includes the graduate degrees.

Because the Census redraws tract lines every ten years, I harmonize every year onto the 2020 boundaries using NHGIS crosswalks, so each tract points to the same ground over time. The rule throughout is to crosswalk the head counts first and compute shares only at the very end. Counts add up; shares do not.

A note on coverage: tract-level attainment exists only for 1990, 2000, and 2012 onward. The long form ran once a decade and there was no annual tract product before the ACS, so the years from 1991 to 1999 and 2001 to 2009 were never collected. Before using the data I ran two checks. The 25 education lines sum back to the total population 25 and over, and my population-weighted BA+ comes to 13.6 percent in 1990 and 17.1 percent in 2000, both matching the published Census figures.

3 Spatial Patterns and Dispersion

The clearest way into the spatial question is to look at the map first. Figure 1 shows BA+ attainment by tract in 1990, 2000, 2012, and 2024, all on the same color scale, so a shade means

the same thing in every panel.

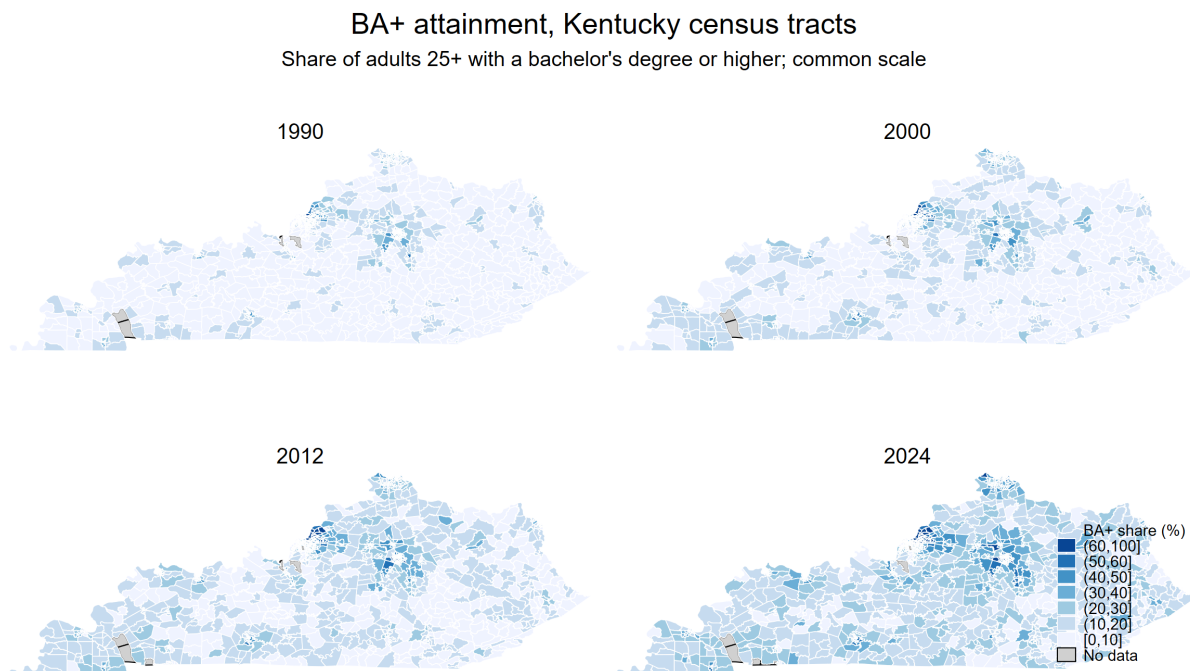


Figure 1: BA+ attainment by census tract, 1990–2024. Darker tracts have a higher share of adults 25 and older with a bachelor’s degree or higher; the color scale is common across panels. Grey tracts are uninhabited or water. Source: decennial census (1990, 2000) and ACS 5-year (2012, 2024); author’s calculations.

Two things happen at once across the panels. The whole state gets darker over time, so attainment rose almost everywhere in Kentucky over these 34 years. But the darker tracts do not spread out. They cluster in three places and stay there: Lexington in the center, the Louisville area, and the Northern Kentucky suburbs across the river from Cincinnati. The south-central and eastern parts of the state, the Appalachian region, stay pale in every panel. They rose a little, but they never catch up. So the gains are uneven. A few metro areas pulled ahead and kept pulling, while rural and Appalachian Kentucky stayed near the bottom the whole time.

The grey tracts in far western Kentucky are uninhabited or water, not missing data. And the map shows where educated adults live, not where they went to school, so a dark suburb means degree-holders live there, which may be because they moved in rather than because the area produced them.

The map shows the gap as geography. The next step is to put a number on it. Figure 2 measures

the spread each year as the gap between the 90th- and 10th-percentile tracts in BA+ share.

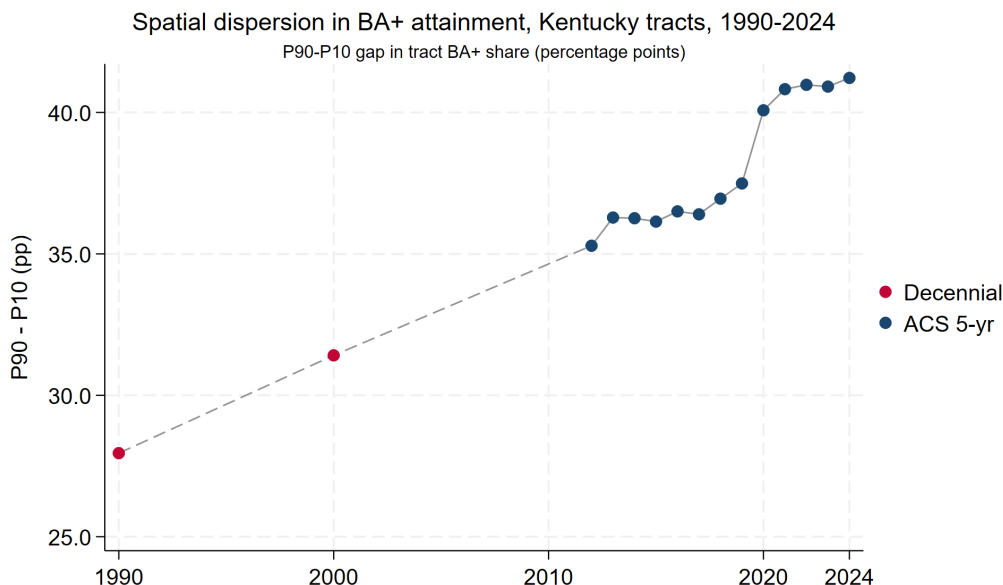


Figure 2: Spatial dispersion in BA+ attainment, Kentucky tracts, 1990–2024. Each point is the gap between the 90th- and 10th-percentile tracts in BA+ share (percentage points). Red points are decennial census years; blue points are ACS 5-year estimates; the dashed segment connects years with no tract-level data. Source: author’s calculations.

That gap widened the whole way, from 27.9 points in 1990 to 31.4 in 2000, 35.3 in 2012, and 41.2 in 2024. By this measure Kentucky’s most- and least-educated places are further apart than at any point in the series. This is *divergence*.

But there is a twist here. Measured against the rising average instead of in raw points, the spread tightened. The coefficient of variation fell from 0.89 in 1990 to 0.64 in 2024. By that measure the distribution converged.

Both are true, and we can see the reason in the tails. The bottom-decile tract went from 3.8 to 9.5 percent BA+, more than doubling. The top-decile tract went from 31.7 to 50.7 percent, up about 60 percent, or 19 points. So the lowest-attainment tracts grew faster in proportional terms, which pulls the relative spread down, while the highest-attainment tracts gained far more in raw points, about 19 against 6, which pushes the absolute gap up. Everyone got more educated, the bottom gained more proportionally, and the top pulled away in absolute size.

We can pay more attention to the absolute gap. A tract at 9 percent BA+ sitting next to one

at 51 percent is a wide divide whatever the ratio did. A few checks support reading it this way. The bottom is not collapsing: the 10th-percentile tract rose in every period, so this is not a story of rural decline. The decennial-to-ACS seam is that the 1990, 2000, and 2012 points fall on a straight progression with no jump, which is mild evidence that the extra harmonization step for the older years did not distort them. The one stretch to read with care is 2019 to 2021, which overlaps the pandemic-era ACS windows, though the 34-year trend does not depend on that segment.

4 The Divergence

The dispersion figure tells us the gap between high- and low-attainment tracts has grown. But it does not tell us what kind of gap. The spread could come from whole counties differing from one another (Fayette county being a more-educated county than Owsley county, or from tracts differing inside the same county, a college-town tract being next to a rural one. Those point to different stories, so it is worth splitting the total spread into the two pieces.

To do so, we use the law of total variance as a tool. We take the total spread in BA+ across all tracts and break it into a between-county part and a within-county part:

$$\underbrace{\frac{1}{W} \sum_i w_i (x_i - \bar{x})^2}_{\text{Total}} = \underbrace{\frac{1}{W} \sum_i w_i (\bar{x}_c - \bar{x})^2}_{\text{Between county}} + \underbrace{\frac{1}{W} \sum_i w_i (x_i - \bar{x}_c)^2}_{\text{Within county}}$$

where x_i is tract i 's BA+ share, w_i its adult population, $\bar{x}$ the state population-weighted mean, $\bar{x}_c$ the mean of the county that tract i is in, and W the total adult population. The between-county part is telling us how far each county is from the state and the within-county part is telling us how far each tract is from its own county. The between share is then simply

$$\text{between share} = \frac{\text{Between}}{\text{Total}}.$$

I weight every tract by its adult population to match the dispersion measure, and I divide all three quantities by the total population so that between plus within equals total exactly. The code checks that identity every year, and it holds. Table 1 and Figure 3 show the result.

Year	Between	Within	Total	Between share
1990	45.0	96.9	142.0	0.317
2000	67.7	118.5	186.2	0.364
2012	89.9	139.1	229.0	0.392
2024	119.1	157.2	276.4	0.431

Table 1: Decomposition of BA+ variance across Kentucky tracts, selected years. Values are the population-weighted variance of tract BA+ share (percentage points squared). The between share is the between-county part divided by the total.

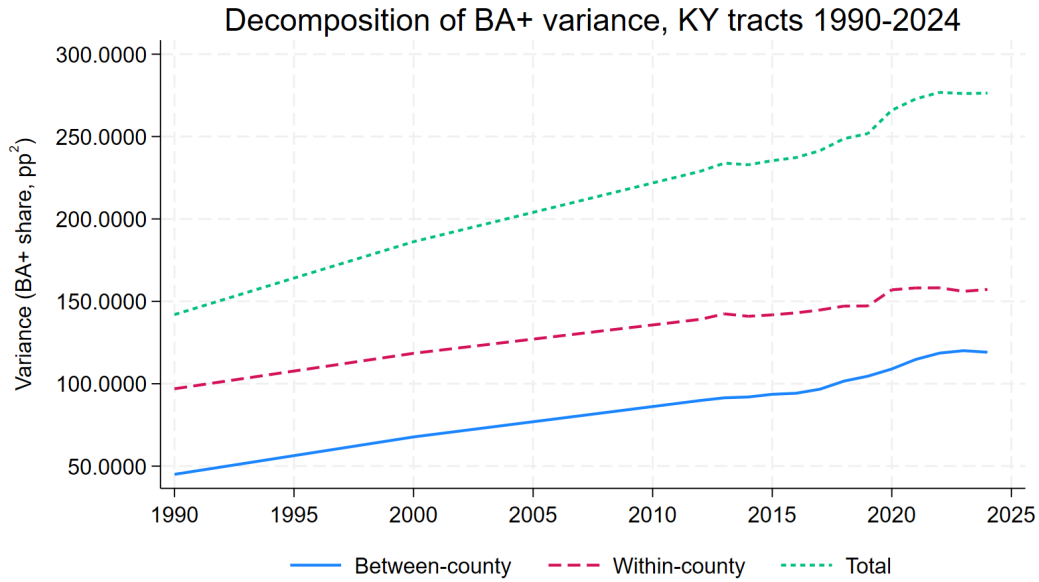


Figure 3: Between-county, within-county, and total variance of tract BA+ share, Kentucky, 1990–2024. The total is the sum of the other two lines in every year. Source: author’s calculations.

Three things worth noting here. First, the total spread roughly doubled, from 142 in 1990 to 276 in 2024. Neighborhoods were already unequal in 1990, and by 2024 they are far more spread out. This is the same divergence the dispersion chart showed, now in variance terms.

Second, within-county spread is the larger piece in every year. In 1990 it was 68 percent of the total; in 2024 it is still 57 percent. Most of the gap in where educated adults live is inside counties, not across them: the college town versus its rural hinterland, the suburb versus the urban core.

Third, the between-county piece is growing faster. Its share rose from 32 percent in 1990 to 43 percent in 2024, climbing consistently the whole way. Put another way, between-county differences account for about 55 percent of the total increase in spread over the period, even though they

started as the smaller share. So the divergence is both kinds at once, with the regional channel doing more of the new growth.

One more thing to read with care, again, is 2019 to 2021, where the within-county line jumps. That window overlaps the pandemic-era ACS disruptions, so the timing of that step should not be read too literally. The longer trend does not depend on it.

The result points to a new question. If the gap between counties is the part that keeps growing, the natural thing to ask is why some counties pull away. That is a question about where graduates are produced and whether they stay, which is the brain-drain extension.

5 Brain Drain

The dispersion and decomposition both point to counties pulling apart. That raises the question of when a county's educated share grows, is it because the county is producing graduates who stay, or because it is pulling in degree-holders from somewhere else? Brain drain is the first of these failing. A place that produces graduates but does not hold onto them is exporting its educated young people.

Answering this cleanly takes two kinds of data. The first is resident attainment, a stock: the share of adults living in a county who hold a degree, and how it grew. I have built this. The second is credential production, a flow: how many graduates a place produces and where they end up. I have not built this yet. The brain-drain signal is the gap between the two. A county that produces graduates but shows weak resident-attainment growth is losing them.

I rolled the full 1990 to 2024 tract panel up to the county level, all 120 Kentucky counties. I computed each county's rate by summing the population counts first and then dividing, which gives the true population-weighted county figure. For each county I have the BA+ share at four benchmark years and the total change across the span. This is the resident half of the comparison.

The county pattern matches the tract. Growth is concentrated in metro and suburban counties and weakest in Appalachia. Table 2 lists the five fastest-growing counties: the Cincinnati suburbs, the Louisville ring, and the Lexington area. Fayette, Lexington itself, are highest in level, rising

from 30.6 percent in 1990 to 48.3 percent in 2024. At the other end, growth was slowest or negative in Eastern Kentucky and Appalachian counties such as Jackson, McCreary, Lee, Elliott, Wolfe, and Carroll. Owsley actually fell, from 9.8 to 7.1 percent.

Table 2: Fastest-growing Kentucky counties, BA+ share, 1990–2024. Change in percentage points; the five largest gains among 120 counties.

County	BA+ growth, 1990–2024 (pp)
Campbell	+24.8
Oldham	+23.5
Woodford	+21.3
Kenton	+20.4
Boone	+20.4

The average county grew 9.9 points, but that is an unweighted average across counties, so it describes the typical county, not the state. Kentucky as a whole grew more than that, because the large counties grew more.

The resident side cannot, by itself, separate brain gain from brain drain. A county can raise its educated share either by producing graduates who stay or by importing adults who already hold degrees. The suburban winners are almost certainly the second kind. Oldham, Boone, and the rings around Cincinnati and Louisville are not minting degrees; they are attracting degree-holders who commute to city jobs. A fast-growing suburb of that sort is brain gain by migration, which is the opposite of what the brain-drain question is after. Telling the two apart is what we can do with the production data does.

6 Summary

Kentucky grew more educated between 1990 and 2024, but the gains were uneven. Attainment rose almost everywhere but are concentrated in a few metro areas, Lexington, Louisville, and Northern Kentucky, with Appalachia near the bottom throughout. The absolute gap between the most- and least-educated tracts widened from 27.9 to 41.2 points, and the decomposition shows the divergence is part regional and part neighborhood-level, with the regional piece growing faster, so whole counties are pulling apart. The county rollup takes that finding toward a brain-drain question the attainment data alone cannot fully answer, which is the next piece of work.

Data and Code Availability

Everything related to this project, including the raw and cleaned data files, R and Stata code, analyses, figures, and tables shown here, is available in the [GitHub repository](#). Anyone with access can rerun the pipeline from the raw files through the final outputs and reproduce these results.